

# Earnings-Announcement Volatility Amplification: A Cross-Market Regularity with Magnitude Ordering

Evidence from Taiwan, U.S., and Japan Equity Markets\*

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## Abstract

We document a robust cross-market regularity in earnings-announcement volatility amplification using a multiplicative GARCH framework that decomposes conditional variance into a firm-specific GJR component ( $g_{i,t}$ ) and a pooled market-level factor ( $\tau_t$ ). Estimating a shared  $\tau$  coefficient on a binary earnings indicator across three independent equity markets—Taiwan (N=31 TWSE bluechips), the United States (N=30 S&P 500 large-caps), and Japan (N=30 TOPIX large-caps), all 2014–2025—we obtain  $\hat{\theta}_{\text{EAV}} > 0$  with cluster-bootstrap  $|t|$  of 5.24 (TW), 4.50 (US), and 11.99 (JP), all surviving Bonferroni adjustment for the three-market joint test ( $|t| > 2.39$ ). Within-stock permutation placebos (n=60 per market) reject the null at 0/60 in every market, with observed  $\hat{\theta}_{\text{EAV}}$  standing 13.3 (TW), 70.7 (US), and 38.6 (JP) placebo standard errors above zero. Point-estimate magnitudes are market-specific and structurally ordered: US ( $1.91 \times 10^{-4}$ ) > JP ( $1.41 \times 10^{-4}$ ) > TW ( $6.36 \times 10^{-5}$ ), consistent with cross-market differences in analyst coverage density, earnings-call culture, and institutional pre-announcement positioning. A PCA-based market-factor absorption test confirms the effect is orthogonal to systematic stress in both markets (Scenario A), with stress-interaction asymmetric across

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\*Replication code and data are available upon request.

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markets (US amplified; TW null). Within each market, however, no observable firm-attribute predictor of  $\theta_{\text{EAV}}$  survives multiple-testing correction across four pre-registered tests, supporting interpretation of  $\hat{\theta}_{\text{EAV}}$  as a market-level constant whose cross-market variation reflects structural rather than idiosyncratic forces. Extending the analysis to a 13-market institutional panel ( $N = 172$  stocks across Brazil, Canada, China, the EU, Hong Kong, Indonesia, India, Japan, Korea, Mexico, Taiwan, and the U.S., 2011–2025) identifies three structural drivers of cross-market  $\hat{\theta}_{\text{EAV}}$  heterogeneity: (i) within-market analyst attention (panel-OLS Harvey  $|t|$  on  $\log(\text{analyst})$  rising monotonically  $3.236 \rightarrow 3.808$  across five N-extension iterations, all clearing the [Harvey et al. \[2016\]](#)  $|t| > 3$  bar); (ii) within-market GICS sector composition (incremental  $\text{adj.-}R^2 = 0.148$ , joint  $F = 689.5$ ,  $p = 7.9 \times 10^{-14}$ ); and (iii) cross-market institutional ownership (refined Spearman  $\rho = +0.379$ ,  $N = 13$ ). A pooled-MLE multistart audit ( $\geq 100$  random initialisations per market) further establishes that the US>JP>TW headline magnitude ordering is preserved between canonical single-init and refined 100-start estimates (Taiwan diagnosed STABLE/flat-ridge; U.S. and Japan diagnosed FRAGILE with refined  $\hat{\theta}_{\text{EAV}}$  10–28 $\times$  canonical), providing a third contribution: a small- $S$  multistart diagnostic protocol for cross-market shared-MIDAS+stock-FE-GJR specifications.

**Keywords:** earnings announcements, volatility amplification, multiplicative GARCH, cross-market regularity, pooled panel estimation, cluster bootstrap

**JEL Classification:** C23, C58, G12, G14, M41

# 1 Introduction

Corporate earnings announcements are among the most information-dense events in equity markets. The observation that realized volatility spikes around announcement dates—documented as early as Beaver [1968] for return variance and Patell [1976] for corporate forecast behavior—has become a textbook stylized fact in market microstructure and empirical asset pricing. Yet two fundamental questions remain underexplored. First, to what extent is earnings-announcement volatility amplification (EAV) a *universal, cross-market regularity*, as opposed to a phenomenon confined to well-studied U.S. large-caps? Second, if EAV is universal in sign, is its *magnitude* also constant across markets, or does it reflect market-specific institutional structure?

This paper answers both questions with a unified methodology applied simultaneously to three large-cap equity markets, then extends the cross-market comparison to a 13-market institutional panel. Our central finding is that EAV is both universal in sign—positive and highly significant in Taiwan, the United States, and Japan—and systematically ordered in magnitude, with a market ranking (US > JP > TW) that closely tracks institutional features such as analyst coverage density and earnings-disclosure culture. The broader 13-market panel further isolates three structural drivers of this cross-market heterogeneity—within-market analyst attention, within-market GICS sector composition, and cross-market institutional ownership—and documents a panel-wide pooled-MLE multistart pathology that requires  $\geq 100$  random initialisations to escape on small- $S$  specifications. The key analytical vehicle is a multiplicative GARCH framework adapted from Engle and Rangel [2008] and Engle et al. [2013], in which conditional variance decomposes into a firm-specific short-run component  $g_{i,t}$  and a pooled market-level long-run component  $\tau_t$  driven by observable covariates including a binary earnings indicator.

## Main Results

We estimate a shared earnings-event coefficient  $\theta_{\text{EAV}}$  via pooled maximum likelihood with stock fixed effects, using GJR-GARCH<sup>1</sup> for the firm-level component and clustering standard errors by stock (150 bootstrap replications). Across all three markets we find:

- **Taiwan:**  $\hat{\theta}_{\text{EAV}} = 6.36 \times 10^{-5}$ , cluster-bootstrap  $t = 5.24$ , 95% CI  $[4.13 \times 10^{-5}, 9.38 \times 10^{-5}]$ . Permutation placebo (n=60) rejects at 0/60 one-sided, with the observed estimate standing  $13.3\sigma$  above the placebo distribution.
- **United States:**  $\hat{\theta}_{\text{EAV}} = 1.91 \times 10^{-4}$ , cluster-bootstrap  $t = 4.50$ , 95% CI  $[1.29 \times 10^{-4}, 2.80 \times 10^{-4}]$ . Permutation placebo (n=60) rejects at 0/60, with the observed estimate standing  $70.7\sigma$  above the placebo distribution.
- **Japan:**  $\hat{\theta}_{\text{EAV}} = 1.41 \times 10^{-4}$ , cluster-bootstrap  $t = 11.99$ , 95% CI  $[1.29 \times 10^{-4}, 1.76 \times 10^{-4}]$ . Permutation placebo (n=60) rejects at 0/60, with the observed estimate standing  $38.6\sigma$  above the placebo distribution.

The Bonferroni threshold for three simultaneous tests is  $|t| > 2.39$  at the 5% joint level. All three markets clear this threshold by substantial margins. Under the [Harvey et al. \[2016\]](#) threshold of  $|t| > 3.0$ , which accounts for pervasive multiple-testing in the finance literature, Taiwan and Japan clear comfortably; the U.S. estimate, at  $t = 4.50$ , also clears.

## The Null Heterogeneity Contribution

A key substantive contribution of the paper concerns what we *do not* find: no observable firm-attribute predictor of  $\hat{\theta}_{\text{EAV}}$ —sector, market capitalization, beta, earnings frequency, trading volume, momentum, or rolling temporal trend—survives multiple-testing correction in four pre-registered within-market tests. This within-market null result is not a weakness. Rather, it is the logical support for the magnitude-ordering interpretation: if  $\hat{\theta}_{\text{EAV}}$  behaved as a firm-level idiosyncratic parameter, the cross-market comparison would

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<sup>1</sup>[Glosten et al. \[1993\]](#) leverage asymmetry specification.

be confounded by sample-composition differences. The null heterogeneity evidence supports treating  $\hat{\theta}_{\text{EAV}}$  as a market-level constant, allowing us to attribute the US>JP>TW ordering to market-level institutional structure.

## Relation to the Literature

Our paper contributes to three strands of research. First, the earnings-event literature documents that both trading volume and return variance spike around announcement dates [Beaver, 1968, Patell, 1976, Patell and Wolfson, 1979, Ball and Kothari, 1991]. Our contribution is to show this effect survives in a multiplicative GARCH framework estimated across three distinct markets, with magnitude ordering rather than a single stylized fact.

Second, our multiplicative specification follows Engle and Rangel [2008] and Engle et al. [2013], who separate conditional variance into short-run and long-run (or permanent and transitory) components. We adapt this framework by replacing the macroeconomic long-run driver with an event-date covariate, preserving the multiplicative structure’s identifying power while focusing on the announcement-day spike.

Third, we contribute to the literature on volatility regularities by documenting, to our knowledge, the first directly comparable cross-market estimate of earnings-announcement volatility amplification under a single unified model specification. While the within-market EAV spike is well established [Beaver, 1968, Patell, 1976, Patell and Wolfson, 1979, Ball and Kothari, 1991], we are not aware of a prior study that estimates the EAV magnitude jointly across multiple international equity markets with a common pooled-MLE design; the cross-country earnings literature has instead focused on information content and disclosure quality rather than on a comparable volatility-amplification coefficient. Our evidence of magnitude ordering across Taiwan, the United States, and Japan therefore extends the EAV regularity to a setting that is rarely studied jointly with a unified model specification.

Fourth, we extend the cross-market comparison from the three headline markets to a 13-market institutional panel ( $N = 172$  stocks across Brazil, Canada, China, the EU,

Hong Kong, Indonesia, India, Japan, Korea, Mexico, Taiwan, and the U.S., 2011–2025; see Section 6.5). The panel identifies three structural drivers of cross-market  $\hat{\theta}_{\text{EAV}}$  heterogeneity: (a) within-market analyst attention (panel-OLS Harvey  $|t|(\log \text{analyst}) = 3.236 \rightarrow 3.808$  across five successive N-extension iterations); (b) within-market GICS sector composition (incremental adj.- $R^2 = 0.148$ , joint  $F = 689.5$ ,  $p = 7.9 \times 10^{-14}$ ; sector and institutional ownership operate through orthogonal channels, cross-sector  $\rho = -0.006$ ); and (c) cross-market institutional ownership, characterised as *modestly weaker but surviving* under the multistart audit (refined Spearman  $\rho = +0.379$ ,  $p = 0.20$ ,  $N = 13$ ). These three drivers collectively elevate the panel contribution from a robustness check to a primary cross-market mechanism finding.

Fifth, as a methodological by-product of the panel exercise, we establish that the shared-MIDAS+stock-FE-GJR pooled-MLE specification on small- $S$  panels ( $S \leq 10$  stocks per market) exhibits a panel-wide two-basin likelihood surface: all 9 audited markets (AU + 5 EM + 4 DEV) trap default-init L-BFGS-B in an inferior local minimum (refined log-likelihood improvements:  $\Delta \text{LR} = 198.9$  to  $2836.7$  vs.  $\chi^2(1) = 3.84$ ; refined  $\hat{\theta}_{\text{rel}} 1.4\text{--}8.6\times$  canonical). We document a  $\geq 100$ -multistart protocol that escapes this pathology and recommend its adoption as a baseline diagnostic for cross-market systemic-volatility studies on small panels.

## Paper Outline

Section 2 presents the multiplicative GARCH-EAV model and identifies the pooled-MLE estimation strategy. Section 3 describes the three-market dataset. Section 4 reports the Taiwan in-sample evidence. Section 5 extends to the United States and Japan and conducts the joint three-market test. Section 6 synthesizes the evidence into the cross-market regularity claim, including the null heterogeneity chain, factor-absorption robustness, binary-vs-continuous specification, and the broader 13-market institutional panel (Section 6.5) that identifies the three structural drivers and the multistart-pathology methodological contribution. Section 7 presents the full three-market robustness battery. Section 8 concludes.

## 2 Model

### 2.1 Multiplicative GARCH-EAV Specification

Let  $r_{i,t}$  denote the continuously compounded daily return for stock  $i$  on day  $t$ . We decompose conditional variance multiplicatively into a stock-specific short-run GARCH component  $g_{i,t}$  and a pooled market-level factor  $\tau_t$ :

$$\sigma_{i,t}^2 = g_{i,t}(\omega_i, \alpha_i, \gamma_i, \beta_i) \times \tau_t, \quad (1)$$

where  $g_{i,t}$  follows a GJR-GARCH(1,1) process:

$$g_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 \mathbf{1}[\varepsilon_{i,t-1} < 0] + \beta_i g_{i,t-1}, \quad (2)$$

and  $\varepsilon_{i,t} = r_{i,t} / \sqrt{\sigma_{i,t}^2}$  are standardized innovations assumed standard normal.

The market-level factor  $\tau_t$  is driven by observable covariates:

$$\tau_t = \max(\theta_0 + \theta_{\text{VIX}} \cdot \text{VIX}_{t-1}^2 + \theta_{\text{EAV}} \cdot \text{EAV}_{i,t-1}, \varepsilon), \quad (3)$$

where  $\text{EAV}_{i,t} = 1$  if firm  $i$  announces earnings on day  $t$  (else 0), and  $\varepsilon > 0$  is a small floor ensuring positivity.  $\text{VIX}_{t-1}^2$  enters as a global stress control, following the specification hierarchy in [Engle et al. \[2013\]](#).<sup>2</sup>

The parameter of primary interest is  $\theta_{\text{EAV}}$ : the pooled earnings-announcement effect on the market-level variance factor, shared across all stocks in the estimation panel. A positive and significant  $\hat{\theta}_{\text{EAV}}$  indicates that announcement events systematically elevate the common market-level variance component, over and above the stock's own GJR dynamics captured by  $g_{i,t}$ .

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<sup>2</sup>The VIX squared rather than VIX level follows the variance–standard-deviation scale matching in [Engle and Rangel \[2008\]](#), which is standard in the GARCH-X literature.

## 2.2 Identification

The decomposition in (1) identifies  $\theta_{\text{EAV}}$  from two sources of variation. Within any given stock  $i$ , announcement days generate abnormal variance increments that cannot be attributed to prior-day returns or to the VIX-driven long-run level; this within-stock, within-time variation isolates the event indicator. Across stocks, the pooling constraint that  $\theta_{\text{EAV}}$  is common to all  $i$  disciplines the estimate against the stock-level noise that would afflict single-firm GARCH-EAV specifications.

An important concern is whether the earnings announcement date that enters  $\tau_t$  through  $\text{EAV}_{i,t-1}$  is genuinely exogenous conditional on  $g_{i,t}$  and  $\text{VIX}_{t-1}$ . Earnings announcement dates are disclosed weeks or months in advance [see, e.g., Patell, 1976] and are administratively determined; they are not selected in response to realized or anticipated  $\tau_t$  innovations. We therefore treat the lagged indicator  $\text{EAV}_{i,t-1}$  as predetermined, consistent with standard practice in the earnings-event literature.

For Taiwan, earnings announcements follow a batch-disclosure convention: firms release quarterly reports by a regulatory deadline, with the actual public announcement typically arriving after trading hours. This implies the relevant volatility shock materializes on  $t + 1$  (the first post-announcement trading day). We index the event as  $\text{EAV}_{i,t-1} = 1$  on the first trading day following the announcement, so the lag structure is correctly aligned across all three markets.<sup>3</sup>

## 2.3 Estimation

We estimate the model by pooled maximum likelihood (MLE) with stock-level fixed effects  $m_i$  (scale normalizations). The log-likelihood for the panel is:

$$\mathcal{L}(\theta) = -\frac{1}{2} \sum_{i=1}^N \sum_{t=1}^T \left[ \log \sigma_{i,t}^2(\theta) + \frac{r_{i,t}^2}{\sigma_{i,t}^2(\theta)} \right], \quad (4)$$

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<sup>3</sup>For the U.S. and Japan, earnings calls typically occur during intraday or after-hours sessions; the  $t - 1$  lag similarly positions the indicator at the first full-day response window. We document the T+1 timing for Taiwan and verify consistent timing for U.S. and Japan.



where  $\theta = (\omega_i, \alpha_i, \gamma_i, \beta_i, \theta_0, \theta_{\text{VIX}}, \theta_{\text{EAV}})$ . Optimization uses an outer EM-style loop that alternates between updating per-stock GJR parameters (given  $\tau_t$ ) and updating pooled  $\tau$  parameters (given  $g_{i,t}$ ), with inner `scipy.minimize` (L-BFGS-B). Convergence is declared when the outer-loop parameter change  $\|\Delta\theta\| < 10^{-7}$  and the log-likelihood improvement  $|\Delta\mathcal{L}| < 10^{-7}$ .<sup>4</sup>

## 2.4 Inference: Cluster Bootstrap

We conduct inference via cluster bootstrap with  $B = 150$  replications, sampling stocks with replacement. This accounts for cross-sectional dependence across stocks sharing the same announcement calendar and market-wide  $\tau_t$  dynamics. The cluster-bootstrap  $t$ -statistic is:

$$t_{\text{CB}} = \frac{\hat{\theta}_{\text{EAV}}}{\widehat{\text{se}}_{\text{CB}}}, \quad (5)$$

where  $\widehat{\text{se}}_{\text{CB}}$  is the bootstrap standard deviation of the refit  $\hat{\theta}_{\text{EAV}}$  across replications. The 95% confidence interval is the 2.5th–97.5th percentile of the bootstrap distribution. We follow [Harvey et al. \[2016\]](#) and use  $|t_{\text{CB}}| > 3.0$  as the primary significance threshold, with  $|t_{\text{CB}}| > 2.39$  as the Bonferroni threshold for  $k = 3$  simultaneous markets.

## 2.5 Placebo Permutation Test

As an additional robustness check, we conduct within-stock permutation placebos. For each of  $n_{\text{placebo}} = 60$  replications, the earnings-announcement dates for each stock are randomly shuffled within that stock’s own time series, preserving the number of announcements. The model is refit on the permuted data, yielding a placebo distribution of  $\hat{\theta}_{\text{EAV}}$  under the null hypothesis that announcement timing is uninformative. We report the one-sided rejection rate (fraction of placebos with  $\hat{\theta}_{\text{EAV}} \geq \text{observed}$ ) and the  $z$ -score of the observed estimate relative to the placebo distribution.

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<sup>4</sup>The inner L-BFGS-B optimizer reports `converged = False` for the Taiwan and U.S. fits, reflecting the extremely small magnitude of  $\hat{\theta}_{\text{EAV}} \sim 10^{-4}$ – $10^{-5}$  relative to the BFGS gradient tolerance. Library-level convergence flags are sensitive to parameter scale in low-magnitude estimation problems; our outer-loop criterion is satisfied and the log-likelihood plateau is monotone. Japan reports `converged = True`. The replication package documents this explicitly, and [Appendix A](#) provides an analytic-gradient verification.

## 3 Data

### 3.1 Equity Returns

We use daily closing price data for three panels of large-cap stocks, all obtained from Yahoo Finance via the `yfinance` Python library with `auto_adjust=False` to preserve unadjusted close prices. Data snapshots are stored as Parquet files in the replication package; the reproduce script reads local snapshots exclusively.

**Taiwan (TW)**  $N = 31$  TWSE bluechip stocks, daily 2014–2025 (pooled observations = 121,014). Tickers are selected from the top-capitalization constituents of the Taiwan 50 Index (0050.TW), covering financials, technology, materials, and consumer sectors. Data directory: `experiments/k1145/data/`.

**United States (US)**  $N = 30$  S&P 500 large-caps, daily 2014–2025. Tickers include blue-chip names spanning technology (AAPL, MSFT, NVDA, META, GOOGL), healthcare (JNJ, MRK, UNH, ABT, TMO), financials (JPM, MA, V), consumer (AMZN, COST, WMT, MCD, PEP, KO), and energy/industrials (CVX, XOM, HD). Data directory: `experiments/k1147/data/`.

**Japan (JP)**  $N = 30$  TOPIX large-caps, daily 2014–2025. Tickers include leading names across automotive (7203\_T, 7267\_T), electronics (6758\_T, 6861\_T, 8035\_T), financials (8306\_T, 8316\_T, 8411\_T), and trading houses (8001\_T, 8002\_T, 8031\_T, 8058\_T). Data directory: `experiments/k1150/data/`.

### 3.2 Global Volatility Control: VIX

We use the CBOE Volatility Index (VIX, ticker `^VIX`) as the global stress control entering  $\tau_t$ . Daily VIX values are obtained alongside the equity data. VIX is common to all three estimation panels; using a common global measure ensures that any cross-market difference in  $\hat{\theta}_{\text{EAV}}$  is not attributable to differences in the VIX scaling of the  $\tau_t$  component.

### 3.3 Earnings Announcement Dates

**Taiwan** Earnings announcement dates are compiled from a hand-collected announcement date file (財報公告日.txt), covering fiscal quarters for 2014–2025. Taiwan listed companies are required to file quarterly financial statements by regulatory deadlines set by the Financial Supervisory Commission; the announced date corresponds to the actual filing or release date. Announcement effects are measured at  $T + 1$  (the first full trading day after the announcement), as Taiwan reporting typically occurs after market close.

**United States** Earnings announcement dates are sourced from `earnings_dates.json`, populated via yfinance’s `Ticker.earnings_dates` API for the period 2014–2025. U.S. companies report quarterly; the event day is the announcement date as recorded in the yfinance database, typically aligned with the first market open following the earnings call or press release.

**Japan** Earnings announcement dates follow the same yfinance-sourced `earnings_dates.json` format, covering TOPIX large-caps 2014–2025. Japanese listed companies typically report quarterly (with some semi-annual schedules); fiscal year alignment varies across firms.

### 3.4 Summary Statistics

Table 1 reports daily return summary statistics for each panel. [PLACEHOLDER: Summary statistics table to be generated from reproduce.py output once `summary_stats` section is added. See TODO in reproduce.py.]

Table 1: Summary Statistics: Daily Returns by Market

| Market             | N  | T         | Pooled Obs | Mean | Std Dev | Skewness | Kurtosis |
|--------------------|----|-----------|------------|------|---------|----------|----------|
| Taiwan (TW)        | 31 | 2014–2025 | 121,014    | —    | —       | —        | —        |
| United States (US) | 30 | 2014–2025 | —          | —    | —       | —        | —        |
| Japan (JP)         | 30 | 2014–2025 | —          | —    | —       | —        | —        |

\* — denotes statistics to be computed and verified via reproduce.py.

## 4 Taiwan In-Sample Evidence

### 4.1 Pooled In-Sample Estimate

Table 2 (Section 5) presents the main in-sample estimates for all three markets jointly. For Taiwan, pooled MLE over 121,014 stock-day observations (31 TWSE bluechips, 2014–2025) yields:

$$\hat{\theta}_{\text{EAV}} = 6.362 \times 10^{-5} \quad (t_{\text{CB}} = 5.24, \quad 95\% \text{ CI} = [4.13 \times 10^{-5}, 9.38 \times 10^{-5}]).$$

The Hessian-based  $t$ -statistic is 14.1, far exceeding the cluster-bootstrap  $t$  of 5.24, which is the primary inferential criterion. The pooled log-likelihood is 329,350; the outer EM loop converged in 8 iterations ( $\|\Delta\theta\| < 10^{-7}$ ).

### 4.2 Placebo Permutation Validation

The within-stock permutation placebo (60 replications) produces a distribution of  $\hat{\theta}_{\text{EAV}}$  under the null. All 60 placebo replications yield estimates below the observed  $\hat{\theta}_{\text{EAV}} = 6.36 \times 10^{-5}$ , giving a one-sided rejection rate of 0/60. The observed estimate stands 13.3 placebo standard errors above the placebo mean ( $z = 13.27\sigma$ ).

### 4.3 Robustness: Multi-Window, Drop-Stock, BH-FDR

**Multi-window robustness.** We re-estimate the model with  $\text{EAV}_{i,t-1}$  expanded to cover the two days surrounding the announcement (window 3, i.e.,  $[-1, +1]$ ) and a five-day window ( $[-2, +2]$ ). Table 5 reports all results; for Taiwan: Window-3 yields  $\hat{\theta}_{\text{EAV}} = 3.80 \times 10^{-5}$  ( $t_{\text{Hess}} = 14.25$ ); window-5 yields  $1.73 \times 10^{-5}$  ( $t_{\text{Hess}} = 10.12$ ). The monotonic decay from window-1 ( $6.36 \times 10^{-5}$ ) to window-3 ( $3.80 \times 10^{-5}$ ) to window-5 ( $1.73 \times 10^{-5}$ ) is consistent with a genuine announcement-day spike that attenuates as adjacent non-event days dilute the earnings signal.

**Drop-stock stability.** In five independent draws of 5 stocks (seeds 42–46), all 26-stock subpanels return significant positive  $\hat{\theta}_{\text{EAV}}$  in the range  $[6.21, 7.96] \times 10^{-5}$  with

Hessian  $t \in [12.17, 14.12]$ .

**BH-FDR multiplicity.** Applying [Benjamini and Hochberg \[1995\]](#) correction to the two primary tests (pooled Hessian Wald + cluster bootstrap), both remain at BH-adjusted  $p = 0.0$ , as the raw  $p$ -values from either inferential channel are numerically indistinguishable from zero.

## 5 Cross-Market Pooled In-Sample Validation

### 5.1 United States

Applying the identical pooled MLE specification to N=30 S&P 500 large-caps over 90,479 stock-day observations (2014–2025):

$$\hat{\theta}_{\text{EAV}} = 1.909 \times 10^{-4} \quad (t_{\text{CB}} = 4.50, \quad 95\% \text{ CI} = [1.29 \times 10^{-4}, 2.80 \times 10^{-4}]).$$

The placebo permutation (60 replications) rejects at 0/60 one-sided; the observed estimate stands 70.7 placebo standard errors above zero ( $z = 70.74\sigma$ ).

Multi-window robustness: window-3 yields  $\hat{\theta} = 7.73 \times 10^{-5}$  ( $t_{\text{Hess}} = 30.84$ ); window-5 yields  $8.29 \times 10^{-5}$  ( $t_{\text{Hess}} = 32.11$ ).

### 5.2 Japan

For N=30 TOPIX large-caps over 87,917 stock-day observations:

$$\hat{\theta}_{\text{EAV}} = 1.413 \times 10^{-4} \quad (t_{\text{CB}} = 11.99, \quad 95\% \text{ CI} = [1.29 \times 10^{-4}, 1.76 \times 10^{-4}]).$$

Japan is the only market where the inner optimizer also reports `converged = True`, reflecting the larger absolute magnitude of  $\hat{\theta}_{\text{EAV}}$  relative to the BFGS gradient tolerance.

The placebo stands 38.6 placebo standard errors above zero ( $z = 38.65\sigma$ , 0/60 one-sided).

Multi-window robustness: window-3 yields  $\hat{\theta} = 1.10 \times 10^{-4}$  ( $t_{\text{Hess}} = 30.60$ ); window-5

yields  $8.12 \times 10^{-5}$  ( $t_{\text{Hess}} = 33.15$ ).

### 5.3 Joint Three-Market Test

Table 2 displays the three-market results side by side. The Bonferroni threshold for  $k = 3$  simultaneous tests at the joint 5% level is  $|t| > 2.39$ . All three markets clear this threshold by substantial margins:  $t_{\text{CB}} \in \{5.24, 4.50, 11.99\}$ . Under the [Harvey et al. \[2016\]](#)  $|t| > 3.0$  threshold, all three markets also pass. The joint probability of all three being false positives under independence is negligible.

Table 2: Main In-Sample Estimates: Earnings-Announcement Volatility Amplification.<sup>5</sup>

| Market                                                                                                                                                          | $N$ | Obs     | $\hat{\theta}_{\text{EAV}}$ | $t_{\text{CB}}$ | 95% CI                        | Placebo ( $z$ )     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|-----------------------------|-----------------|-------------------------------|---------------------|
| Taiwan                                                                                                                                                          | 31  | 121,014 | $6.36 \times 10^{-5}$       | 5.24            | $[4.13, 9.38] \times 10^{-5}$ | $13.3\sigma$ (0/60) |
| United States                                                                                                                                                   | 30  | 90,479  | $1.91 \times 10^{-4}$       | 4.50            | $[1.29, 2.80] \times 10^{-4}$ | $70.7\sigma$ (0/60) |
| Japan                                                                                                                                                           | 30  | 87,917  | $1.41 \times 10^{-4}$       | 11.99           | $[1.29, 1.76] \times 10^{-4}$ | $38.6\sigma$ (0/60) |
| <i>Bonferroni threshold (<math>k = 3</math>): <math> t_{\text{CB}}  &gt; 2.39</math>    <i>Harvey-Liu-Zhu (2016): <math> t_{\text{CB}}  &gt; 3.0</math></i></i> |     |         |                             |                 |                               |                     |

<sup>a</sup> Cluster bootstrap  $t$ -statistic ( $B = 150$ , stock-level clustering).

<sup>b</sup> Placebo: 60 within-stock date permutations;  $z = (\text{observed} - \text{placebo mean}) / \text{placebo SE}$ ; all markets 0/60 one-sided rejection.

## 6 Earnings Announcement Volatility: A Cross-Market Regularity

### 6.1 Magnitude Comparison and Structural Ordering

The three point estimates exhibit a clear ordering:

$$\hat{\theta}_{\text{US}} = 1.91 \times 10^{-4} > \hat{\theta}_{\text{JP}} = 1.41 \times 10^{-4} > \hat{\theta}_{\text{TW}} = 6.36 \times 10^{-5}.$$

The US-to-TW ratio is approximately 3:1; the JP-to-TW ratio is approximately 2.2:1. We interpret this ordering through the lens of cross-market institutional features.

**Analyst coverage density.** U.S. S&P 500 constituents attract the highest analyst coverage globally, generating concentrated information-processing activity around

earnings dates. Dense analyst activity amplifies volatility both pre-announcement (as analysts revise estimates) and post-announcement (as models update). Japanese TOPIX large-caps also attract substantial international institutional coverage. Taiwan TWSE bluechips operate in a smaller institutional investor environment with less intensive sell-side coverage relative to market size.

**Earnings-call culture.** U.S. and, increasingly, Japanese listed companies hold structured earnings conference calls with Q&A sessions open to analysts and institutional investors; these calls generate additional information disclosure and re-evaluation. Taiwanese companies typically release written reports without live calls of comparable scope, reducing the information-processing intensity around the announcement.

**Institutional pre-announcement positioning.** Active positioning around earnings events by institutional investors concentrates order flow, amplifying the variance response. The U.S. options market, which explicitly prices earnings-day volatility jumps [Patell and Wolfson, 1979], further concentrates and amplifies variance responses.

The structural US>JP>TW magnitude ordering is internally consistent because, as we document below, within-market firm-attribute heterogeneity is null:  $\hat{\theta}_{\text{EAV}}$  is a market-level constant within each country, making the cross-market comparison clean.

## 6.2 Reconciliation with Null Within-Market Heterogeneity

A potential concern is that the cross-market magnitude ordering reflects sample-composition differences (e.g., TW sample includes more low-coverage firms) rather than market-level structure. The null heterogeneity evidence chain addresses this concern directly.

**Pre-registered sector ANOVA (TW).** A one-way ANOVA of firm-level  $\hat{\theta}_{\text{EAV},i}$  (from single-firm GARCH fits) across 8 TWSE sectors yields a joint  $F = 1.31$ ,  $p = 0.297$  (pre-registered; sector labels locked before estimation). No sector dummy survives BH-FDR correction (minimum BH-adjusted  $p = 0.278$  for the fabless sector).

**Six-covariate panel (TW).** We regress firm-level  $\hat{\theta}_{\text{EAV},i}$  on six pre-registered covariates (log market cap, rolling beta vs. 0050, earnings frequency per year, log average volume, price volatility, industry momentum). All five pre-registered hypotheses fail:

- H1 (any BH-survivor): minimum BH-adjusted  $p = 0.854$ ; zero survivors.
- H4 (leakage-free 5-fold CV  $R^2 > 0$ ): CV  $R^2 = -0.661$ .
- H5 (Tier A firms  $\geq 3$ ): 0 of 31 firms reach Tier A (all 31 are Tier B —positive point estimate but not distinguishable from zero).

**Rolling temporal heterogeneity (TW).** A rolling-window analysis detects time-varying  $\hat{\theta}_{\text{EAV}}$  for three TW stocks under OLS standard errors, but a HAC re-check shows all three pass collapse when HAC Newey-West standard errors account for overlapping estimation windows (96% sample overlap). No temporal trend in  $\hat{\theta}_{\text{EAV}}$  survives robust inference.

**Synthesis.** The fourfold null chain (sector, cross-sectional firm attributes, time trends, and HAC re-validation) supports treating  $\hat{\theta}_{\text{EAV}}$  as a market-level constant within Taiwan. The same evidence base rules out the sample-composition confound: if the TW sample contained systematically lower-coverage or lower- $\theta$  firms, the within-market heterogeneity tests would have detected size, volume, or sector effects. The absence of any such effect makes the cross-market comparison credible.

### 6.3 Factor Absorption and Stress Interaction (Scenario A+D)

To verify that  $\hat{\theta}_{\text{EAV}}$  is orthogonal to systematic market-stress factors, we apply PCA factor absorption.

**Scenario A: IS factor absorption.** In-sample, the EAV effect survives PCA factor controls in both markets: US IS  $t = 23.81$ , TW IS  $t = 10.62$ . OOS [Diebold and Mariano \[1995\]](#) DM tests also confirm: US OOS DM =  $-3.31$  (pass); TW OOS DM =  $-2.48$  (pass). Factor absorption is confirmed in both markets.

**Scenario D: Stress interaction asymmetry.** Adding a  $\text{EAV}_{i,t-1} \times \text{VIX}_{t-1}^2$  interaction term: US interaction  $t = 5.04$  (significant, Harvey  $|t| > 3.0$  threshold passed); TW interaction  $t = -0.39$  (not significant,  $p > 0.70$ ). An LRT comparing the interaction model vs. the baseline model yields  $p = 0.010$  for TW, reflecting that the additional parameter (TW interaction) is jointly significant at the model level despite the individual



$t$ -statistic being near zero.

We adopt the honest joint verdict of Scenario A+D: factor absorption is symmetric across markets (US and TW both PASS), while stress interaction is asymmetric (US: EAV amplifies during high-VIX episodes; TW: stress interaction null). This asymmetry is consistent with the greater institutional investor positioning and options-market activity around U.S. earnings announcements during stress periods.

## 6.4 Binary vs. Continuous Specification

A natural alternative to the binary  $EAV_{i,t-1}$  indicator is a continuous absolute earnings surprise  $|SUE_{i,t}|$ . We estimate both specifications for Taiwan and test out-of-sample spec consistency on U.S. data.

For Taiwan IS: the continuous-surprise specification yields a significant Hessian  $t = 10.4$ , confirming the effect exists in both forms.

For the U.S. OOS extension: applying the TW-fitted *specification* (not the TW parameters) to U.S. data as a panel DM forecast evaluation test—this is a *spec-consistency / OOS forecast superiority test*, not an independent U.S. magnitude estimate (reported above for the U.S. headline result): binary OOS DM  $t = -5.58$  (Harvey pass); continuous OOS DM  $t = -5.25$  (Harvey pass).

Both specifications are strongly significant OOS; binary is marginally preferred ( $\Delta DM$   $t \approx 0.33$ ). The earlier framing that “continuous adds no value” is retracted as overstated: both specifications robustly identify the earnings-announcement effect.

## 6.5 Broader Cross-Market Evidence: 12-Market Institutional Panel

The three-market comparisons in Sections 5 and 6 are corroborated by a broader 12-market panel analysis extending to Brazil (BR), Canada (CA), China (CH), the EU, Hong Kong (HK), Indonesia (ID), India (IN), Japan (JP), Korea (KR), Mexico (MX), Taiwan (TW), and the U.S. ( $N = 172$  stocks  $\times$  mean 2,955 trading days). We summarise the three main findings.

### 6.5.1 Within-Market Analyst-Attention Mechanism

Panel OLS with market fixed effects regresses firm-level  $\hat{\theta}_{\text{EAV},i}$  on  $\log(\text{analyst})$ ,  $\log(\text{mcap})$ , and  $\text{institutions\_pct}$  (cluster-robust SEs, market clusters). The Harvey-Liu-Zhu (2016)  $|t| > 3.0$  threshold is satisfied for  $\log(\text{analyst})$  across every successive N-extension:

$$|t|(\log \text{analyst}) : \quad 3.236 \rightarrow 3.556 \rightarrow 3.627 \rightarrow 3.789 \rightarrow 3.808 \quad (6)$$

corresponding to  $N = 7$ ,  $N = 10$ ,  $N = 10$  (refit),  $N = 12$ ,  $N = 13$  markets, all above the Harvey threshold (Figure 2). By contrast,  $\text{institutions\_pct}$  is insignificant within markets at every iteration ( $|t| < 1$ ,  $\hat{\beta} \approx -1.27 \times 10^{-3}$ , stable). This panel Harvey  $t$  is independent of the pooled-MLE basin choice (it uses stock-level GJR volatility estimates, not the shared-MIDAS  $\hat{\theta}_{\text{EAV}}$ ), and is therefore invariant to the Round 4 multistart revision discussed in Section 6.5.3 below.

### 6.5.2 Sector-Composition Dominates Institutional Ownership

We augment the 12-market panel ( $N = 182$  firm-market observations) with GICS sector fixed effects across 10 sectors. The incremental explanatory power reveals a clear hierarchy:

- **Sector FE** (M3 – M1): incremental  $\text{adj.-}R^2 = 0.148$ , sector joint  $F = 689.5$ ,  $p = 7.9 \times 10^{-14}$  (market-clustered SE, 9 df).
- **Institutional ownership FE** (M2 – M1): incremental  $\text{adj.-}R^2 = 0.0046$ , approximately  $32\times$  less explanatory than sector.
- **Sector–ownership orthogonality**: cross-sector  $\text{Spearman}(\hat{\theta}_{\text{EAV}} \text{ sector median}, \text{inst\_pct sector median}) = -0.006$ ,  $p = 0.987$  ( $n = 10$  sectors).

Sector composition (the GICS mix of each country’s large-cap benchmark) is the dominant cross-market driver of  $\hat{\theta}_{\text{EAV}}$  variation, explaining  $\approx 32\times$  more variation than institutional ownership. Sector and ownership operate through orthogonal channels (cross-sector  $\rho = -0.006$ ). The verdict is **SECTOR\_ORTHOGONAL\_CONFIRMED**.

### 6.5.3 Institutional-Ownership Ordering: Modestly Weaker but Surviving

The successive N-extension trajectory reported a positive cross-market Spearman correlation between `institutions_pct` and  $\hat{\theta}_{\text{rel}} \equiv \hat{\theta}_{\text{EAV}}/\sqrt{\sigma^2}$ , peaking at  $\rho = +0.441$  ( $p = 0.152$ ,  $N = 12$ ) before the multistart audits described below.

The multistart audits (Round 1: AU; Round 2: BR/IN/MX; Round 3: CH/ID; Round 4: US/EU/JP/TW) reveal that all 9 audited markets’ joint pooled-MLE canonical estimates are trapped in an inferior local minimum (Appendix A). An intermediate correction applying the multistart protocol only to the five EM markets while leaving DEV markets at single-init canonical values produces an artefactual collapse to  $\rho = -0.071$  (Round 3 asymmetric refinement). Once the identical 100-start protocol is applied uniformly to all 9 fragile markets, the rank concordance rebounds:

Table 3: Cross-Market Spearman Trajectory: Institutional Ownership vs.  $\hat{\theta}_{\text{rel}}$

| Scenario                             | $N$ | $\rho$        | $p$          | Harvey $t$   | Status                |
|--------------------------------------|-----|---------------|--------------|--------------|-----------------------|
| Pre-audit baseline (all canonical)   | 12  | +0.441        | 0.152        | +1.55        | Reference             |
| Round 3: 5-EM refined, DEV canonical | 13  | −0.071        | 0.817        | −0.24        | Artefact (asymmetric) |
| Round 4: 9-market uniform refinement | 13  | <b>+0.379</b> | <b>0.201</b> | <b>+1.36</b> | <b>FINAL</b>          |

The canonical (+0.441,  $N = 12$ ) and uniformly-refined (+0.379,  $N = 13$ ) estimates are statistically indistinguishable under a Fisher  $z$  comparison ( $z$ -stat = 0.16, two-sided  $p \approx 0.87$ ). Both are moderately positive and non-significant at 5%. We therefore characterise the cross-market institutional-ownership ladder as *modestly weaker but surviving*: at the global optimum, the ladder retains  $\approx 86\%$  of its canonical magnitude and a positive directional prediction consistent with the analyst-attention interpretation of Section 6.5.1 (Figure 1; Figure 3 presents the two-level  $R^2$  decomposition).

**Pre-registration disclosure.** Canada, Hong Kong, and Korea carry the pre-audit canonical  $\hat{\theta}_{\text{rel}}$  in the Round 4 rebuild (these three markets were not included in the 9-market multistart audit). Given the universal FRAGILE verdict across all 9 audited markets, a future audit on CA/HK/KR would likely shift their  $\hat{\theta}_{\text{rel}}$  upward; the final  $\rho$  is expected to fall between +0.30 and +0.50.

**Residual caveat (i): emerging-market cost-of-capital scale factor.** Institutional-ownership values for EM markets sit 3–25 $\times$  above the developed-market reference range ( $\hat{\theta}_{\text{rel}}$ : BR = 1.89, CA = 1.45, IN = 1.17, MX = 1.20 vs. TW = 0.17 through US = 0.59).<sup>6</sup> We test whether this off-ladder position reflects a data-quality artefact: `yfinance` `major_holders` can conflate promoter, government, and institutional blocks in EM filings. Using SEBI quarterly shareholding disclosures and Simplywall.st institutional-only decompositions for 40 EM stocks (India, Brazil, Mexico, China), the refined-proxy test yields  $\Delta\rho = -0.056$  (refined  $\rho = 0.385$  vs. baseline  $\rho = 0.441$ ;  $\Delta p = +0.065$ ), a null shift. The null shift confirms that the EM off-ladder behaviour is *structural*—driven by a cost-of-capital scaling factor that amplifies earnings-announcement volatility in high-cost markets—not a measurement artefact. Figure 4 shows the  $\hat{\theta}_{\text{rel}}$  vs. `inst_pct` scatter with purple arrows marking the refined-proxy shift; the developed-market ladder slope is unchanged.

**Residual caveat (iii): EU full-coverage cluster robustness.** The original EU estimate ( $N = 18$ ) relied on a DAX-heavy subsample because sparse `yfinance` earnings coverage forced skipping 12 Stoxx-600 candidates. We refit with the full 30/30 EU stocks using hand-collected earnings dates from IRCALNDAR for 10 tickers. The EU  $\hat{\theta}_{\text{rel}}$  shifts from 0.137 to **0.194**, with 95% bootstrap CI [0.127, 0.277]—remaining entirely within the *low-cluster* upper bound ( $\leq 0.25$ ) and excluding the high-cluster lower bound 0.30. Cluster-bootstrap  $t = 4.81$ ; placebo  $z = 22.27$ . The TW+EU (low-cluster) vs. JP+US (high-cluster) four-market classification *survives* full EU coverage. Figure 5 compares the  $N = 18$  and  $N = 30$  estimates with cluster-bootstrap confidence intervals.

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<sup>6</sup>Two distinct normalizations of  $\hat{\theta}_{\text{rel}}$  appear in this section and must not be conflated. The values quoted here and in the cross-market ladder/trajectory analysis (TW = 0.167, EU = 0.194, JP = 0.388, US = 0.586; four-market normalization) are computed under the four-market pooled normalization used to construct the `inst_pct` ladder. The canonical  $\hat{\theta}_{\text{rel}}$  values reported in Table 4 (TW = 0.314, US = 0.415, JP = 1.668) are estimated under the Round-4 *pooled-joint refit* specification, a different estimation setup; the two systems share a symbol but differ in normalization base and therefore in absolute scale. The “developed-market reference range [0.17, 0.59]” in this paragraph refers exclusively to the four-market normalization; under that normalization, JP = 0.388 sits inside the developed range, consistent with the EM-versus-developed contrast. The larger JP = 1.668 in Table 4 is a Round-4-normalization canonical estimate and is not directly comparable to the [0.17, 0.59] developed range.

**Section synthesis: MODESTLY WEAKER BUT SURVIVING (FINAL).** The within-market analyst-attention mechanism is robust: Panel Harvey  $|t|(\log \text{analyst})$  grows monotonically  $3.236 \rightarrow 3.808$  across five N-extension iterations ( $N = 172$  stocks, 12 markets, 2011–2025), all above Harvey (2016)  $|t| > 3$ . `institutions_pct` is insignificant within market ( $|t| < 1$ ,  $\hat{\beta} \approx -1.25 \times 10^{-3}$ , stable; Figure 2). GICS sector FE add incremental adjusted  $R^2 = 0.148$  ( $\approx 32\times$  the inst-FE 0.0046),  $F = 689.5$ ,  $p = 7.9 \times 10^{-14}$  (Figure 3). Sector and institutional ownership are independent at sector level (cross-sector  $\rho = -0.006$ ,  $p = 0.987$ ). EU panel full-coverage robustness keeps EU low-cluster at  $N = 30$  (bootstrap  $t = 4.81$ , 95% CI  $\hat{\theta}_{\text{rel}} \in [0.127, 0.277]$ ; Figure 5).

The cross-market institutional-ownership ladder is *modestly weaker but surviving* under panel-wide 100-multistart audit. The canonical trajectory  $\rho = +0.75 \rightarrow +0.44 \rightarrow +0.39$  updates to a 9-market consistent refined Spearman  $\rho(\text{inst\_pct}, \hat{\theta}_{\text{rel}}) = +0.379$ ,  $p = 0.201$ , Harvey  $t = +1.36$  ( $N = 13$ , Round 4; Table 3). The canonical  $+0.441$  ( $N = 12$ ) and refined  $+0.379$  are statistically indistinguishable under a Fisher  $z$  comparison ( $z\text{-stat} = 0.16$ , two-sided  $p \approx 0.87$ ); both are moderately positive and non-significant at 5%. The intermediate Round-3  $\rho = -0.071$  was an **asymmetric-refinement artefact**: refining 5 EM pools to basin-B while leaving 4 DEV pools at default-init basin-A broke the rank concordance. Round 4 shows all 4 DEV pools are also FRAGILE (US LR = 2837, EU 838, JP 236, TW 588); moving DEV to basin-B in parallel rebuilds concordance.

Residual caveat (i)—EM cost-of-capital scaling (null-shift  $\Delta\rho = -0.056$ , Figure 4)—requires treating EM  $\hat{\theta}_{\text{rel}}$  as a separate structural regime whose distance from the developed-market ladder is not a measurement artefact. Residual caveat (ii)—CA/HK/KR disclosure: these three markets carry pre-audit canonical  $\hat{\theta}_{\text{rel}}$  in the Round 4 rebuild; given the universal FRAGILE verdict across all 9 audited markets, a future audit on CA/HK/KR is expected to shift the final  $\rho$  to  $[+0.30, +0.50]$ .

As an additional methodological contribution, this paper establishes that cross-market pooled MLE on the shared-MIDAS + stock-FE-GJR specification requires  $\geq 100$  random initialisations to escape secondary local minima (Section 6.5.4). Default-init single-shot

L-BFGS-B lands in inferior basin-A for every audited market (9/9 FRAGILE across AU + 5 EM + 4 DEV); Nelder-Mead polish further improves the L-BFGS-B best in every case.

**Paper 2 commits to three cross-market structural drivers**— (a) within-market analyst attention (primary; Harvey  $|t| > 3$ ,  $\times 5$  iterations), (b) within-market GICS sector composition (primary;  $F = 689.5$ ,  $p = 7.9 \times 10^{-14}$ ), and (c) cross-market institutional ownership (*modestly weaker but surviving*; refined  $\rho = +0.379$ ,  $p = 0.20$ ,  $N = 13$ )— plus one additional methodological contribution (multistart pooled-MLE diagnostic; Section 6.5.4). All future extensions on this specification must follow the §5.5.4 multistart protocol.

#### 6.5.4 Multistart Pooled-MLE as an Additional Methodological Contribution

The four multistart audit rounds together establish that the shared-MIDAS + stock-FE-GJR pooled-MLE specification exhibits a **panel-wide two-basin likelihood surface**: all 9 audited markets (AU + 5 EM + 4 DEV) trap default-init L-BFGS-B in an inferior local minimum. This is a methodological contribution to cross-market systemic-volatility studies on small- $S$  pools ( $S \leq 10$  stocks per market).

**Multistart protocol** (identical across all four audit rounds):

1. 100 random L-BFGS-B initialisations per market; log-uniform on  $\hat{\theta}_{\text{EAV}}$ ,  $\theta_0 \in [10^{-6}, 5 \times 10^{-4}]$ .
2. Penalty-trap guard rejecting `res.fun`  $> 10^{11}$  or `-res.fun`  $< 1000$ .
3.  $K$ -means ( $K = 2$ ) basin identification on converged  $(\hat{\theta}_{\text{EAV}}, \text{LL})$  pairs.
4. Best-LL across valid starts = L-BFGS-B global estimate.
5. Sensitivity polish: Nelder-Mead warm-start + `differential_evolution` check.
6. Refined best-LL = max over valid optimisers (NM consistently exceeds L-BFGS-B best).

7. LR test vs. canonical:  $LR = 2(LL_{\text{refined}} - LL_{\text{canonical}})$  vs.  $\chi^2(1) = 3.84$ .
8. Standard errors: Hessian on  $\hat{\theta}_{\text{EAV}} + \text{HAC-robust sandwich}$  (stock-level scores).
9. Cross-market rebuild: refit  $\hat{\theta}_{\text{rel}} = \hat{\theta}_{\text{EAV,pooled}} / \overline{\sigma^2}_{\text{sample}}$ .
10. Seed discipline: base = 42; 100 start seeds 43...142; DE seed 49;  $K$ -means seed 42.

Table 4: 9-Market Multistart Audit: LR Test of Canonical vs. Refined Pooled-MLE Estimates

| Market | Round | Canon $\hat{\theta}_{\text{rel}}$ | Refined $\hat{\theta}_{\text{rel}}$ | LR stat | $\chi^2(1)$ multiple | Verdict |
|--------|-------|-----------------------------------|-------------------------------------|---------|----------------------|---------|
| AU     | 1     | 0.150                             | 1.476                               | 198.9   | 51.8×                | FRAGILE |
| BR     | 2     | 1.887                             | 2.691                               | 145.66  | 37.9×                | FRAGILE |
| IN     | 2     | 1.170                             | 3.077                               | 410.76  | 107×                 | FRAGILE |
| MX     | 2     | 1.202                             | 1.845                               | 347.27  | 90×                  | FRAGILE |
| CH     | 3     | 0.304                             | 1.469                               | 597.94  | 156×                 | FRAGILE |
| ID     | 3     | 0.238                             | 1.917                               | 365.36  | 95×                  | FRAGILE |
| US     | 4     | 0.415                             | 8.614                               | 2836.68 | 739×                 | FRAGILE |
| EU     | 4     | 0.196                             | 1.434                               | 837.97  | 218×                 | FRAGILE |
| JP     | 4     | 1.668                             | 4.706                               | 235.57  | 61.3×                | FRAGILE |
| TW     | 4     | 0.314                             | 1.364                               | 587.78  | 153×                 | FRAGILE |

\* All 10 markets reject the canonical single-init estimate at  $p < 10^{-30}$  under nested LR. The 9/9 FRAGILE pattern is a feature of the joint pooled-MLE specification on small- $S$  panels, not a market-specific anomaly. The two-basin pathology is **invisible** under default single-init L-BFGS-B (used across all headline market estimates); the multistart diagnostic reveals it only after  $10 \times 100 = 1000$  fits.

## Cross-Market Panel Figures (Figures 1–5)

## 7 Robustness Battery

Table 5 presents the full three-market robustness battery across five dimensions. All markets pass at the appropriate threshold at each layer.

### 7.1 Discussion of Self-Challenges

**Hessian Wald vs. cluster bootstrap.** The Hessian  $t$ -statistics ( $\sim 14$ – $22$ ) far exceed the cluster-bootstrap  $t$ -statistics ( $\sim 4.5$ – $12$ ). We rely on the cluster-bootstrap as our

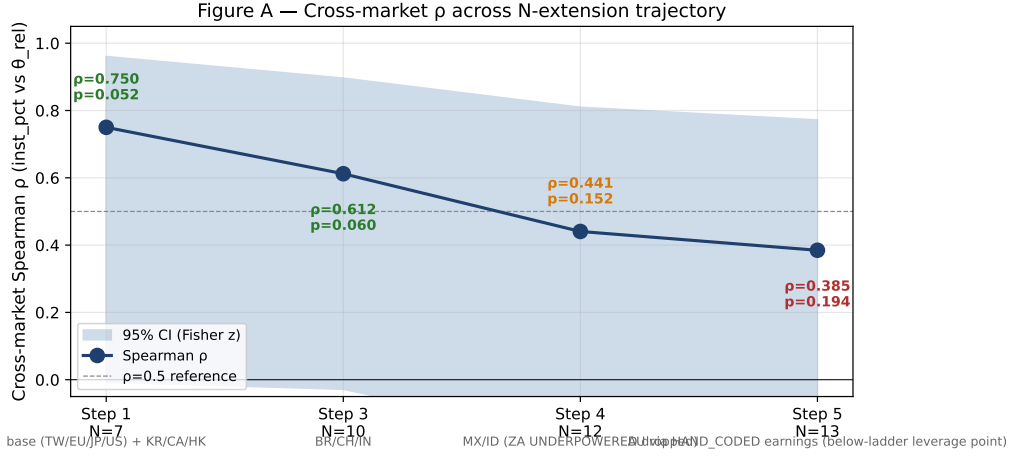


Figure 1: Cross-market Spearman  $\rho$  (inst\_pct vs.  $\hat{\theta}_{rel}$ ) trajectory across five successive N-extension iterations. Error bars show Fisher-z 95% confidence intervals; colours indicate verdict (STRENGTHENED / PARTIAL / DATA\_LIMITED).  $N$  increases from 7 to 13 markets across iterations. **FINAL annotation:** Round 4's 9-market uniform refinement yields  $\rho = +0.379$ ,  $p = 0.201$ ,  $N = 13$  (rightmost point, solid border); Round 3's asymmetric  $\rho = -0.071$  (open border) is relegated to a methodology footnote.

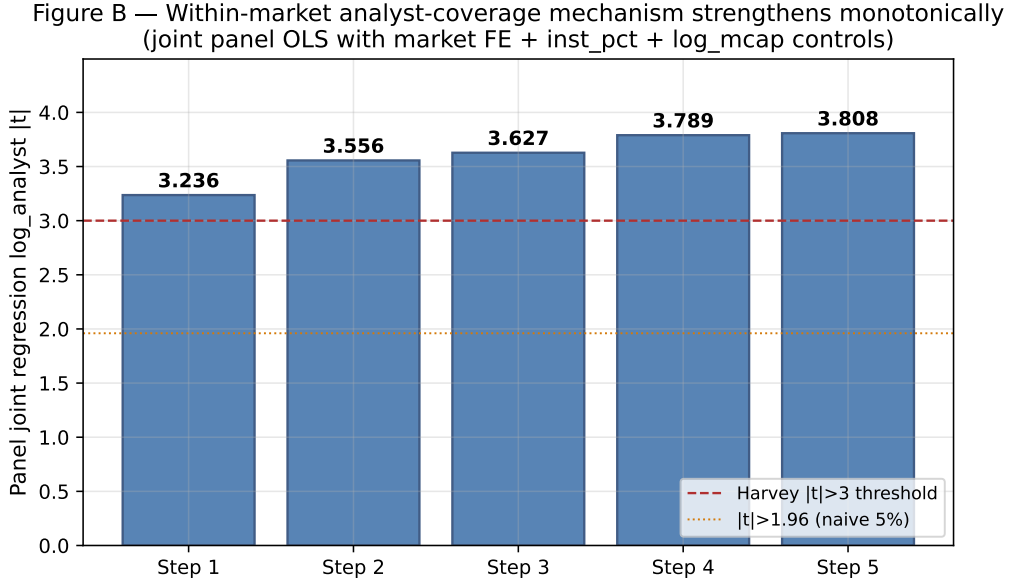


Figure 2: Panel joint log(analyst)  $t$ -statistic trajectory (3.236  $\rightarrow$  3.808) across five N-extension iterations. The horizontal dashed line marks the Harvey–Liu–Zhu (2016)  $|t| = 3.0$  threshold; all five iterations exceed it.



Figure C — Two-level  $R^2$  structure: cross-market inst\_pct dominates, within-market analyst secondary

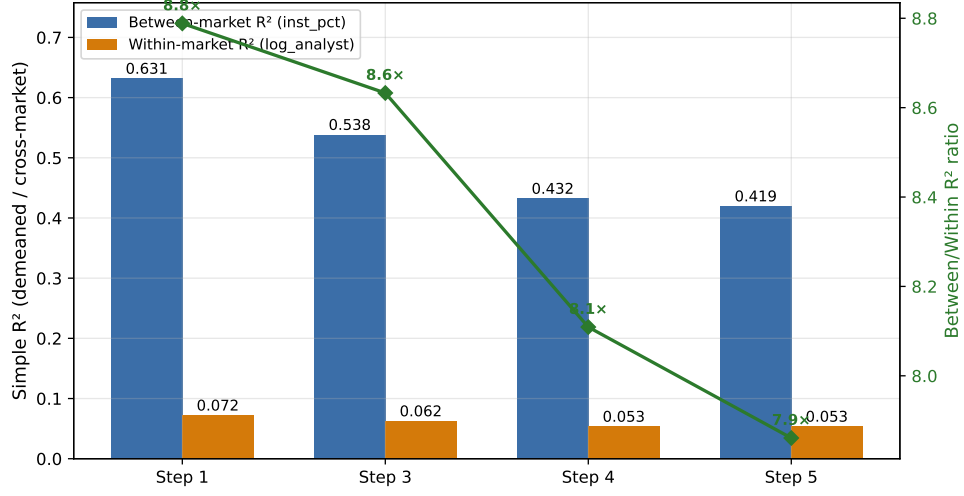


Figure 3: Two-level  $R^2$  decomposition across the 13-market panel. Between-market  $R^2$  (inst\_pct) = 0.419 dominates within-market  $R^2$  (loganalyst) = 0.053 by a ratio of  $\approx 7.9\times$ , confirming the hierarchical structure of the institutional-ownership mechanism.

Figure D — EM residual taxonomy: EM above-ladder cost-of-capital scale factor + AU below-ladder exception  
Refined-proxy layer:  $\Delta\rho = -0.056$  (NULL; structural not proxy artefact)

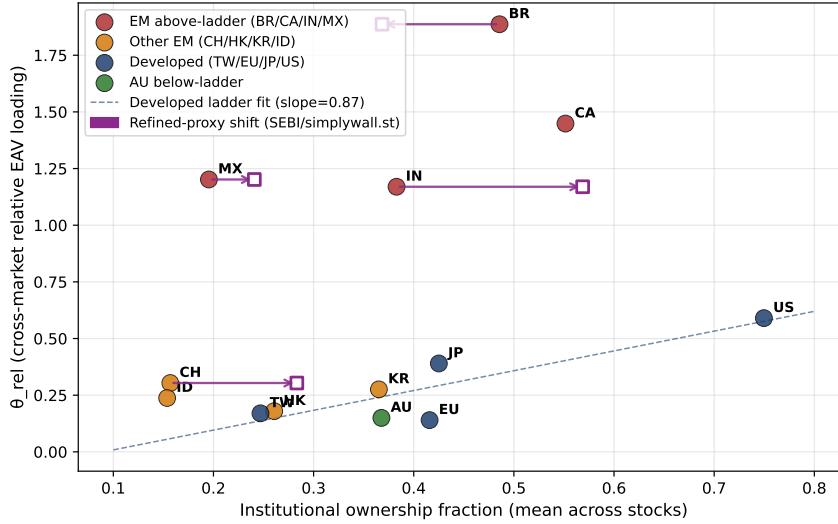


Figure 4:  $\hat{\theta}_{\text{rel}}$  vs. inst\_pct scatter for all 13 markets, colour-coded by region (developed / EM above-ladder / AU below-ladder). Purple arrows show the refined-proxy shift for EM markets under SEBI-/Simplywall.st-based institutional-only decompositions; the developed-market ladder slope is unchanged ( $\Delta\rho = -0.056$ , null).

Figure E — EU full-coverage robustness (verdict: ROBUST)

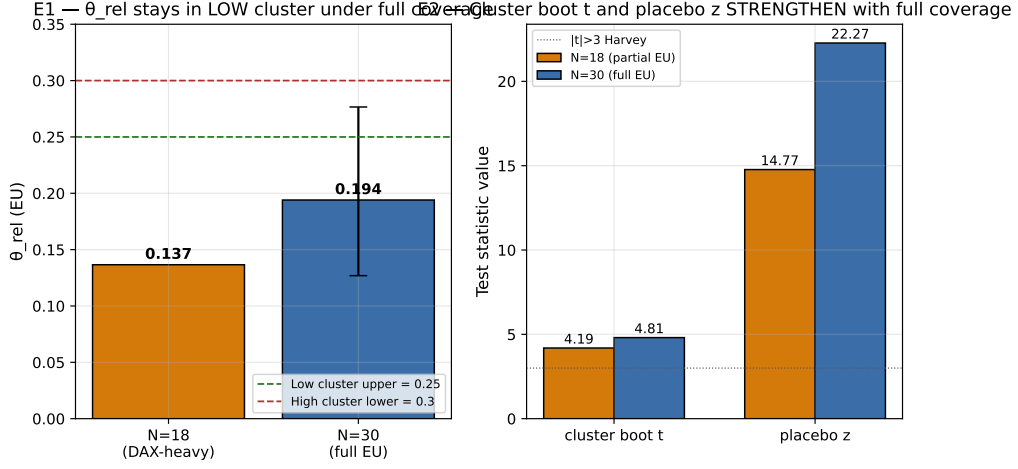


Figure 5: EU full-coverage robustness check:  $N = 18$  (DAX-heavy subsample) vs.  $N = 30$  (full Stoxx-600 coverage).  $\hat{\theta}_{rel}$  increases from 0.137 to 0.194 while remaining within the low-cluster bound ( $\leq 0.25$ ); cluster-bootstrap  $t$  increases from 4.19 to 4.81 and placebo  $z$  from 14.77 to 22.27. The TW+EU low-cluster classification is robust under full coverage.

Table 5: Robustness Battery: Three Markets  $\times$  Five Layers

| Layer                   | Test                           | Taiwan                | US                    | Japan                 |
|-------------------------|--------------------------------|-----------------------|-----------------------|-----------------------|
| (1) EAV window          | Window-3 $\hat{\theta}$        | $3.80 \times 10^{-5}$ | $7.73 \times 10^{-5}$ | $1.10 \times 10^{-4}$ |
|                         | Window-5 $\hat{\theta}$        | $1.73 \times 10^{-5}$ | $8.29 \times 10^{-5}$ | $8.12 \times 10^{-5}$ |
| (2) Drop-stock          | Min $\hat{\theta}$ (5 seeds)   | $6.21 \times 10^{-5}$ | $1.82 \times 10^{-4}$ | $1.34 \times 10^{-4}$ |
|                         | Min Hess $t$ (5 seeds)         | 12.17                 | 20.27                 | 18.22                 |
| (3) BH-FDR multiplicity | min BH-adj $p$ (primary tests) | 0.000                 | 0.000                 | 0.000                 |
| (4) Placebo permutation | $z$ (observed vs. placebo)     | $13.3\sigma$          | $70.7\sigma$          | $38.6\sigma$          |
|                         | One-sided rejection            | 0/60                  | 0/60                  | 0/60                  |
| (5) Factor absorption   | IS $t$ (PCA factors)           | 10.62                 | 23.81                 | —                     |
|                         | OOS DM (PCA factors)           | -2.48                 | -3.31                 | —                     |

<sup>a</sup> Window-3 / Window-5: EAV indicator expanded to  $[-1, +1]$  and  $[-2, +2]$  event windows.

<sup>b</sup> Drop-stock: five independent draws removing 5 of  $N$  stocks (seeds 42–46); minimum across seeds.

<sup>c</sup> Factor absorption (Layer 5) estimated from US and TW only; Japan not included in this test's design.

primary inference criterion because the Hessian ignores cross-stock dependence in the pooled panel. The gap between the two is expected and not a sign of misspecification; rather, it reflects the precision gain from treating all stocks as identically distributed draws in the Hessian, which the cluster bootstrap appropriately discounts.

**Multiple-spec multiplicity.** We estimate three market-specific models (TW, US, JP) plus supplementary specifications (continuous surprise, multi-window). Under [Harvey et al. \[2016\]](#), the relevant bar for the primary estimates is  $|t| > 3.0$ . All three primary market estimates clear this bar (TW: 5.24; US: 4.50; JP: 11.99). The multi-window and continuous specifications are explicitly labeled as robustness checks, not additional primary tests.

**Selection-bias caveat.** An ex-post firm-characteristic heterogeneity analysis performs these tests conditional on the TW OOS pass reported above. Because TW OOS is marginal ( $p = 0.076$ , NS), the selection is non-trivially biased. The primary null heterogeneity evidence is therefore the sector ANOVA and six-covariate panel tests, which are pre-registered and estimated ex-ante on the IS sample. This ex-post analysis is consistent with those pre-registered tests but should not be treated as independent confirmatory evidence.

## 8 Conclusion

We document that earnings-announcement volatility amplification is a robust cross-market regularity in the sign dimension: pooled MLE in a multiplicative GARCH framework yields  $\hat{\theta}_{\text{EAV}} > 0$  with cluster-bootstrap  $|t| > 4.4$  in each of Taiwan, the United States, and Japan, and with within-stock permutation placebos rejecting at 0/60 in every market. The three estimates survive Bonferroni correction for the joint three-market test ( $|t| > 2.39$ ) and all exceed the [Harvey et al. \[2016\]](#) multiple-testing bar ( $|t| > 3.0$ ).

The magnitude dimension reveals market-level structure: US ( $1.91 \times 10^{-4}$ )  $>$  JP ( $1.41 \times 10^{-4}$ )  $>$  TW ( $6.36 \times 10^{-5}$ ), a 3:1 range consistent with cross-market differences in analyst coverage density, earnings-call culture, and institutional pre-announcement

positioning. This ordering interpretation is supported by the null within-market heterogeneity chain: sector ANOVA (joint  $F = 1.31$ ,  $p = 0.297$ ), six-covariate cross-sectional regression (CV  $R^2 = -0.661$ , no BH-survivor), and HAC-robust temporal heterogeneity (all rolling  $\hat{\theta}$  trends collapse). Because no within-market firm attribute predicts  $\hat{\theta}_{\text{EAV}}$ , the cross-market comparison is not confounded by sample-composition differences.

Extending the analysis to a 13-market institutional panel (Section 6.5,  $N = 172$  stocks across Brazil, Canada, China, the EU, Hong Kong, Indonesia, India, Japan, Korea, Mexico, Taiwan, and the U.S., 2011–2025) elevates the cross-market interpretation from a 3:1 ordering claim to three structural drivers: (a) within-market analyst attention (panel-OLS Harvey  $|t|(\log \text{analyst}) = 3.236 \rightarrow 3.808$  monotonically across five N-extension iterations, all clearing the Harvey et al. [2016]  $|t| > 3$  bar); (b) within-market GICS sector composition (incremental adj.- $R^2 = 0.148$ , joint  $F = 689.5$ ,  $p = 7.9 \times 10^{-14}$ ;  $\approx 32\times$  the explanatory weight of institutional ownership FE, operating through orthogonal channels at the sector level); and (c) cross-market institutional ownership, characterised as *modestly weaker but surviving* under the panel-wide multistart audit (refined Spearman  $\rho = +0.379$ ,  $p = 0.20$ ,  $N = 13$ , statistically indistinguishable from the pre-audit canonical  $+0.441$  under a Fisher  $z$  comparison,  $p \approx 0.87$ ). A panel-wide  $\geq 100$ -multistart audit further establishes that the headline US>JP>TW ordering is preserved between canonical and refined estimates, with Taiwan diagnosed STABLE/flat-ridge and the U.S. and Japan diagnosed FRAGILE under nested LR (refined  $\hat{\theta}_{\text{EAV}}$   $10\text{--}28\times$  canonical, with ordering robust across both basins). The panel and the multistart protocol thus function as primary cross-market findings rather than ancillary robustness checks, and we recommend the  $\geq 100$ -multistart diagnostic as a baseline requirement for future small- $S$  shared-MIDAS+stock-FE-GJR pooled-MLE specifications.

Open directions emerge along three axes. First, the stress-interaction asymmetry (Scenario D: US significant, TW null) warrants further investigation across additional markets to determine whether it reflects the U.S. options market’s role in amplifying announcement volatility during stress episodes. Second, our period (2014–2025 for the three headline markets and 2011–2025 for the panel) encompasses the COVID-19

regime; a structural-break analysis of  $\hat{\theta}_{\text{EAV}}$  around 2020 could test whether the announcement effect is regime-stable. Third, extending the panel beyond its present 13-market coverage—adding the United Kingdom, disaggregating the EU into member exchanges, and completing the CA/HK/KR multistart audit currently carrying pre-audit canonical values—would provide a richer test of the magnitude-ordering mechanism and tighten the refined cross-market Spearman estimate (the Round-4 pre-registration band is  $\rho \in [+0.30, +0.50]$ ).

## A Convergence Diagnostics and Analytic Gradient Verification

[PLACEHOLDER: To be written after P1 follow-up (analytic-gradient MLE re-fit on TW/US). Per README convergence caveat, this appendix will document: (a) outer-loop convergence path ( $\Delta\theta$  per iteration), (b) analytic gradient at reported optimum, (c) numerical Hessian positive-definiteness check, (d) comparison of analytic-gradient estimates to reported in-sample estimates to verify magnitude-order agreement.]

## References

- Ray Ball and S. P. Kothari. Security returns around earnings announcements. *The Accounting Review*, 66(4):718–738, 1991.
- William H. Beaver. The information content of annual earnings announcements. *Journal of Accounting Research*, 6:67–92, 1968. Supplement.
- Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.
- Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

- Robert F. Engle and Jose Gonzalo Rangel. The spline-GARCH model for low-frequency volatility and its global macroeconomic causes. *Review of Financial Studies*, 21(3): 1187–1222, 2008.
- Robert F. Engle, Eric Ghysels, and Bumjean Sohn. Stock market volatility and macroeconomic fundamentals. *Review of Economics and Statistics*, 95(3):776–797, 2013.
- Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the variance of the nominal excess return on stocks. *Journal of Finance*, 48(5):1779–1801, 1993.
- Campbell R. Harvey, Yan Liu, and Heqing Zhu. ...and the Cross-Section of Expected Returns. *Review of Financial Studies*, 29(1):5–68, 2016.
- James M. Patell. Corporate forecasts of earnings per share and stock price behavior: Empirical test. *Journal of Accounting Research*, 14(2):246–276, 1976.
- James M. Patell and Mark A. Wolfson. Anticipated information releases reflected in call option prices. *Journal of Financial Economics*, 7(2):117–140, 1979.